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A REVIEW OF POST ACTIVATION POTENTIATION AND ITS APPLICATION IN STRENGTH AND CONDITIONING

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Power production may be the most important characteristic of sport performance, and may determine the quality of performance.⁴⁹ In this way, any training intervention that has the capability of acutely enhancing power performance has the potential to directly enhance sports performance. Prior to the effective integration of any training intervention into the training of athletes, any such phenomenon needs to be subjected to effective research to determine both its potential for performance enhancement, along with methods of optimal application. One training intervention that has been the focus of an increasing degree of research into its potential power enhancement properties is post activation potentiation (PAP). PAP refers to an excited or sensitive neuromuscular condition following intense loading,⁴³ a situation where power output can be potentially augmented. This potential has led to the development of systems such as complex training,⁹ based around this phenomenon, whereby power performance can potentially be enhanced via the application of a preconditioning exercise, for example a squat.⁹ While an increasing body of research now exists into the application of PAP to performance,²⁶ much of the evidence is still contradictory. This paper attempts to produce an effective summary of the current research, starting with an examination of the scientific underpinning of PAP and also examining its application to the acute enhancement of power performance.

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The coexistence of potentiation and fatigue

PAP involves the use of a preconditioning activity, to potentially augment the power output of a subsequent activity, e.g. the use of a pre-conditioning squat to augment counter movement jump (CMJ) performance. Any conditioning activity, whilst potentially enhancing PAP, will also have a fatiguing effect on skeletal muscle. In this way, any conditioning activity will initiate both potentiating and fatiguing responses,⁸ and despite the contradictory effects of PAP and fatigue "these two presumed mechanisms can coexist".³²

Fatigue can be defined as "an exercise induced reduction in the maximal force capacity of muscle".²⁸ In relation to PAP, fatigue is associated with a reduction in maximum force output on the subsequent effort, caused by the initial exercise. However, in addition to the fatiguing effects of an activity, a brief period of activity can increase both the electrical and mechanical output of a muscle above resting values.¹⁶ In this way, a potentiation effect can be elicited via a preceding exercise, in addition to any fatiguing effects. The net effect on human performance of prior contractile history is therefore the net balance between fatigue inducing processes and the simultaneous processes that result in potentiation.²⁶

Trade off between PAP and fatigue

The time lines of fatigue and PAP will influence the post-stimulus state,⁴² and the optimal recovery window is dependent upon the amplitude and rate of decay of PAP and fatigue. Thus, any PAP inducing activity is going to have a trade off effect on fatigue, and this must be accounted for when planning for PAP, and also when evaluating the effectiveness of PAP in enhancing performance. Peak PAP will occur immediately after the conditioning activity, and while this is the time of peak PAP, it is unlikely to be the time where

peak performance will be evident. This is due to the fact that this will also be the time of peak fatigue. The relative magnitude of PAP and fatigue is dependent upon both the nature of the pre-conditioning exercise and the characteristics of the athlete. Intense activities, whilst evoking greater PAP, will also likely evoke greater fatigue and vice versa. Similarly, the greater the time difference between the conditioning activity and subsequent performance, the greater the recovery from fatigue, but also the greater the decrement in PAP. It is likely that a degree of trial and error will be required in determining the optimal PAP window for an individual athlete, and this needs to be remembered when designing programmes that utilise PAP inducing methodologies. A more detailed discussion of the timelines associated with PAP can be found later in the paper.

Underlying physiological mechanisms of PAP

Although there is general consensus regarding the existence of PAP, the mechanisms underlying it are yet to be fully determined.⁴² Skeletal muscle performance is affected by its contractile history,²⁶ and "twitch potentiation and fatigue in skeletal muscle are two conditions in which force production is affected by the stimulation history".⁴⁰ To date in the literature, two mechanisms have been proposed to explain PAP.¹⁰ The first is based upon the phosphorylation of myosin light chains,¹² while the latter is based upon an increase in neural excitability.²² These reflect the contribution of both physiological and neural influences to the phenomenon of PAP.

Rassier and Macintosh⁴⁰ and Paasuke *et al.*,³⁷ suggest that the mechanism responsible for PAP is the "phosphorylation of the regulatory light chains of myosin". Muscle stimulus causes an increase in sarcoplasmic Ca^{2+} , that activates myosin light chain (MLC) kinase,⁵⁰ which can increase actin-myosin crossbridging, through increasing ATP availability.¹² Additionally, the phosphorylation of the MLC renders the actin-myosin interaction more sensitive to Ca^{2+} released from the sarcoplasmic reticulum.⁴⁴ This could theoretically increase the rate at which myosin cross bridges achieve a force producing state.²⁶ However, this effect is at its peak at low myoplasmic levels, which occurs at low frequency contractions. These low frequency contractions would therefore be of insufficient magnitude to enhance power performance. At high frequency tetanic contractions, increased sensitivity to Ca^{2+} has little or no effect in saturating Ca^{2+} .⁴⁴ This would suggest that PAP's effect would be smallest where motor unit firing rates are likely to be at their highest i.e. during high power outputs.⁴⁴ Tubman *et al.*⁵² examined MLC phosphorylation and post-tetanic potentiation and concluded that it was not the only mechanism contributing to PAP, and PAP is likely to be the result of the interaction between neural and muscular mechanisms.¹² In this way, the physiological mechanisms of PAP highlighted thus far seem to offer limited effectiveness in enhancing strength or power performance, as PAP induced via this mechanism would not be able to elicit an increase high frequency force".⁴⁴

In terms of enhanced neural excitability, a number of neural mechanisms have been proposed by which PAP can potentially enhance power performance. This potential neural basis of PAP is evidenced by a

potentiated reflex response following prior stimulation.⁵¹ This enhanced reflex response can result in enhanced reflex amplitudes relative to constant stimulation intensities, and can indicate some form of synaptic modification at the spinal level.²⁶ This could be due to increased motor neuron pool excitability, variations in neurotransmitter release by afferent terminals or variations in a motor neuron's intrinsic properties.²⁶ The exact mechanism by which these neural influences elicit enhanced performance is not known,¹ but may occur through the recruitment of more motor units, better motor unit synchronisation, a decrease on pre-synaptic inhibition of greater central input to the motor neuron, or a combination of these mechanisms.² Whichever mechanisms or combinations predominate, it could be that PAP results in an enhanced synaptic efficiency between Ia afferent terminals and α motor neurons of the stimulated muscle,²⁶ resulting in an environment that supports enhanced power performance.

Effects on power performance

While the complex mechanisms which are likely to contribute to PAP have yet to be determined, perhaps a more important consideration for strength and conditioning coaches is how PAP can impact power performance, and the aspects of power performance which will be affected by PAP. Peak power is likely the critical factor in sports that require maximal effort single movements such as jumping sprinting etc⁴⁹ and thus any enhancement of power performance through PAP is likely to need to focus on this key variable. Given that power can be defined as $Power = Force \times Velocity$, then any enhancement of performance through PAP will need to impact either force output, the velocity of force output, or both.

In terms of force output,²¹ assert that peak isometric force, and maximum un-resisted shortening velocity cannot be enhanced by PAP and thus contributions to improved performance through PAP cannot be accounted for via these mechanisms. However, studies on twitch potentiation have reported an increase in peak twitch force, and so the impact of PAP on peak force cannot be discounted.

If the impact of PAP of peak force is indeed limited, then any major impact of PAP is likely to be at the velocity end of the power equation, and must therefore impact on the rate of force development and on movement velocity. It is postulated that the rate of force development can be enhanced, even at very high stimulation frequencies.⁴⁴ This enhancement of the rate of force development (RFD), is a key effect of PAP in Type II fibres and hence for explosive force production. This is vital for speed and power performance, as they are characterised by brief maximal efforts, with a higher number of motor units recruited and firing at maximal rates.⁴⁴ In this way PAP will be at its greatest in activities in which the rate of force development is an important component. This affect on velocity of movement was supported by Stone *et al.*⁴⁹ who found a significant increase in peak movement velocity in elite weightlifters using a potentiated sequence of sets, where lower load sets were preceded by heavy sets.

The application of PAP to sports performance

Initial studies on PAP focussed on the twitch characteristics of muscles. A twitch is a brief

contraction of a muscle in response to short (<1ms) electrical stimulation of a nerve.³⁶ In these studies twitch characteristics are measured prior to, and after, muscle activity. These activities have included evoked tetanic contractions,³⁶ sustained maximal voluntary contractions,²¹ or a series of submaximal evoked twitches.³³ These studies have consistently shown a potentiated response or twitch potentiation. These have been characterised by increases in the RFD, a decrease in relaxation time and an increase in peak twitch force.¹² Additionally, studies on skinned mammalian skeletal muscle fibres have consistently shown increased twitch tension, increased rate of tension development and decreased post-stimulus relaxation time.⁴²

Despite their value in determining the potential for PAP, and the underlying mechanisms of PAP, of critical interest to strength and conditioning coaches is the potential of this phenomenon to improve sports performance. While PAP has been consistently demonstrated in lab based experiments, its effect on measures of sports performance are far more equivocal. While a number of studies have demonstrated increased performance via PAP, others have shown no change or even a deterioration in performance. Therefore, a closer investigation of the phenomena of PAP is warranted.

Studies supporting PAP

Young *et al.*,⁵⁶ using strength trained athletes, showed a significant ($p<0.05$) increased jump height of 2.8% in loaded counter movement jumps after the performance of a set of half squats, using a recovery time of 4 minutes. They suggested that a set of heavy resistance work results in a degree of heightened neural stimulation, and this results in an improved performance on a subsequent plyometric or maximal power movement. Guellich & Schmidtbleicher²² also found that speed strength performances of the lower extremities in sprinting and jumping events were considerably improved following maximal voluntary contractions. Baker⁴³ found that performance on a second set of jump squats preceded by a set of heavier squats, resulted in a 17.2% increase in jump height over a first set not preceded by a set of heavy squats, although no statistical evidence was given to support this data. Similarly Radcliffe & Radcliffe,³⁹ found improved standing long jump performance when preceded by heavy resistance work, in this case the power snatch. Chiu *et al.*,⁷ in establishing the effects of PAP on jump squats, utilised 5 sets of 1 repetition to stimulate PAP and demonstrated significant increases in peak power, but only in strength trained athletes. No significant differences were found in untrained athletes or the subject group when taken as a whole. Gourgallis *et al.*²⁰ demonstrated a mean improvement of 2.39 ($p<0.05$) on vertical jump height following 5 sets of half squats for 2 repetitions at 20%, 40%, 60% 80% and 90% 1RM, they also indicated that subjects with greater maximal strength levels in the squat experienced greater potentiation than those with lower maximal strength. Hamada *et al.*²³ showed significant PAP following maximal isometric contractions on the performance of a trained muscle group in triathletes and runners. Rixon *et al.*⁴² found that PAP, (as measured by counter-movement jump performance), could be induced with isometric squats (3 reps of 3 seconds of maximal voluntary isometric contraction in a squat against a fixed resistance with two minutes between repetitions), but not with dynamic squats (1 set

of 3 repetitions at 90% 1RM on a cybex Smith machine). Gilbert *et al.*¹⁹ demonstrated no increase in maximum voluntary contractions but a significant increase in RFD (on an isometric leg press at 90 degree leg angle, with data measured by a strain gauge linked to an analogue to digital converter) ($p=0.016$), after performance of 5 single repetitions of maximal squat exercises of five single repetitions at 100% 1RM. This increase in RFD peaked at 20 minutes post squat. This supports the work of Sale⁴⁴ who suggests that PAP will have its greatest effects on RFD, with peak force remaining unchanged.

McBride *et al.*,³⁵ investigating the effect of PAP on speed, found that athletes ran 0.87% faster in a 40 yard dash, ($p<0.05$) when preceded by a set of heavy squats (1 set or 3 repetitions at 90% of 1 RM). Interestingly, no statistical difference was seen in the split times at 10 or 30 metres. This suggests that PAP has greatest effects at the highest running speeds and is interesting given the relative change from the important of maximal strength to elastic strength as a sprint progresses. Chatzopoulos *et al.*,⁶ investigating the effects of PAP on speed, found no change in performance over 10 metres and 30 metres following 10 sets of single squats at 90% 1RM, after 3 minutes recovery, but did find significant ($p<0.05$) improvements after 5 minutes. Yetter *et al.*⁵⁵ found significant improvements in the 10-20 metre section of a 40 yard sprint following back squats, but not front squats. Again supporting the variations found in the results of other studies, no differences were found at the other section of a 40 yard dash. This suggests that there may be differences in the potentiation potential of differing exercises, and that potentiation can affect different aspects of speed performance to varying degrees. These findings Smith *et al.*⁴⁶ also postulated that PAP may be able to increase running speed, following their findings of enhanced 10 second sprint cycling performance following 10 x 1 repetition 1RM back squats.

Kilduff *et al.*³¹ found a statistical ($p<0.05$) difference between pre and post potentiating performance in both upper body power (bench throws) and lower body power (countermovement jump), but only after a rest period of 8 minutes. In this instance, the potentiating activity was a 3RM bench press for upper body and squat for the lower body. Weber *et al.*⁵⁴ using 5 repetitions of 85% 1 RM demonstrated significant improvement in mean and peak jump height and peak ground reaction force, after a recovery period of 3 minutes. Stone *et al.*⁴⁹ found that manipulating the order of sets into strength-power potentiating complexes by preceding lighter load sets with heavier sets resulted in improvements in peak force, peak power, rate of force development and peak velocity, but only the increase in peak velocity was significant.

Even within the studies demonstrating enhanced performance through PAP, there is a degree of variance between the results, with not all elements of power being enhanced equally. The equivocal nature of the results is clearly demonstrated in French *et al.*'s study,¹⁸ who found significant increases ($p<0.05$) in drop jump height, maximal force production and acceleration impulse after a sequence of maximal isometric knee extensions of 3 repetitions of 3 seconds; yet found no statistical increase in counter-movement jump height. This is despite the fact the countermovement jump height is related to acceleration impulse. Additionally, no EMG differences were found in any measures.

Studies showing no effects on performance

While the previous studies have demonstrated the beneficial effects of PAP on different measures of performance, other studies show no enhancement of performance via PAP methodologies. Mangus *et al.*³⁴ in a study on weightlifters, reported no statistically significant enhancement in vertical jump performance after a set of squats of 1 repetition at 90% 1RM. Despite no statistical difference, they did report a trend ($p < 0.07$) towards enhanced performance. In this protocol it needs to be noted that a single repetition at 90% 1RM may not have been sufficient to maximise PAP, where most studies have used 3 repetitions at this load. However similarly this group did utilise a strength trained subject group, which has been suggested will benefit to a greater degree from PAP.⁸

Ebben *et al.*¹⁵ found no acute enhancement on medicine ball power drop performance subsequent to a preceding set of high load bench press. Jensen and Ebben³⁰ showed no improvement in jumping performance following a set of 5 RM squats, demonstrating a trend to decline at short rest intervals following the pre-conditioning activity, although they found increasing performance with increased rest and suggested that had the recovery period been extended beyond 4 minutes, performance could have been enhanced. Gossen and Sale²¹ found no improvement in isometric peak knee torque post PAP inducement via a 10 second MVC, indeed demonstrating a trend for decline in performance. However, in this study recovery time was 15 seconds, and the authors suggested that a longer recovery time, even at the cost of a diminished PAP, may have proved beneficial. Bazett-Jones⁵ also found no PAP effect on peak force and rate of force development following 3 sets of leg presses at 90% 1RM for 3 repetitions. Again a trend to reduced performance was reported three minutes after the completion of the conditioning exercise. Hrysomallis and Kidgell²⁷ also found that a "heavy dynamic resistive upper body exercise (5 RM bench press) was inadequate in augmenting short-term power (explosive push ups)". Scott and Docherty⁴⁵ found no significant difference in either vertical jump or horizontal jump performance as a consequence of performing a set of 5 squats at 5RM load. Robbins and Docherty⁴³ also found no acute enhancement of countermovement jump performance after the execution of a maximal voluntary isometric contraction of seven seconds.

Hanson *et al.*²⁵ found no enhancement of jump performance following the squat exercise in resistance trained men, at either high velocity low load, or low velocity higher load. In this case it needs to be noted that the resistance levels were set at 40% (8 repetitions) on the first testing day and 80% (4 repetitions) on the second. Neither of these resistances may have been sufficient to induce PAP.²⁵ Similarly Bazett Jones *et al.*⁵ found no improvement in peak force or rate of force development subsequent to 3 sets of 3 repetitions in the leg press, with 90% of 1 RM. However in this instance the authors contend that any potential potentiation effects could have been masked by the fatiguing nature of the protocol. Smith and Fry⁴⁷ found no enhancement of change of force, velocity or power measures on the performance of a 1RM leg extension 7 minutes subsequent to a 10 second MVC. Seven subjects in the study demonstrated elevated regulatory myosin light chain (RLC)

phosphorylation, but showed no increase in performance. Therefore even where RLC phosphorylation is present, this does not guarantee enhanced performance. Interestingly, a further four subjects actually demonstrated a reduced RLC phosphorylation further emphasising the individuality of responses on a physiological as well as performance basis. It should be noted that this study used recreationally trained men, who may not be able to optimally utilise PAP.

Comyns *et al.*¹⁰ found no significant change in CMJ performance following 1 set of 5 back squats at a 5RM load, indeed this study found a significant reduction in performance at 30 seconds and 6 minutes. Comyns *et al.*¹¹ also found significant decreases in drop jump flight time subsequent to 3 repetitions of squats at 65% 80 % and 93%. No significant differences were reported for peak ground reaction force, and reactive strength index at all loads. However they did see a significant increase in leg stiffness at 93% load. Again this study demonstrates the complexity of responses to any PAP inducing activity, and the difficulties in relating a specific physiological change to a change in performance.

General conclusions of studies to date

The results to date are equivocal in terms of the capacity of PAP to enhance performance, the optimal method of inducing PAP and also on the exact mechanisms by which performance can be enhanced through PAP inducing protocols. In this way, while PAP may appear to have potential to enhance performance, its utilisation across all subject groups needs to be treated with caution. Similarly, definitive guidelines as to the optimal application of PAP remain impossible to set out at this time.

An increasing theme of the conclusions of published is that PAP may exist, but that the exact protocols used in the study did not elicit it, and that factors such as different loads, different rest periods, different subjects etc, may have produced different results. While these conclusions are valid, it is important that coaching practices are geared to producing quantifiable performance improvements through scientifically validated methodologies. PAP may indeed be a physiological phenomenon, but unless it can be optimally used to enhance performance on a consistent basis, then constant modification of protocols to utilise it, may in fact be a futile search, especially if any modifications negatively affect the training impact of other training modalities. For example, if rest periods of 8 minutes optimise PAP for a given individual, then utilising PAP modalities that use this rest periods may be counterproductive if the volume load of other elements of the session such as the heavy resistance training are reduced to such a degree to limit their effectiveness. An effective cost benefit analysis should be put in place if PAP methodologies result in a necessary modification of other training stimuli.

Key issues in PAP application

Despite the potential of PAP for enhancing power performance, the logistics of optimising these in coaching situations provides a major challenge. In determining the optimal application of any PAP-inducing activity in the training of athletes, a coach needs key information on a number factors.

These include:

1. The nature of the PAP inducing modality
2. The determination of optimal loads to elicit PAP
3. The determination of the optimal timeframe of PAP
4. The characteristics of the athlete

The nature of PAP inducing modalities

PAP has been demonstrated to be produced via both MVC's or dynamic resisted exercises such as squats, although other studies have demonstrated no significant effects on performance with both modalities. PAP seems to be a phenomenon that is independent of contraction type, isometric, concentric or eccentric⁴⁹ and at present, no methodology appears to have an advantage. However, Hilfiker *et al.*²⁵ suggest that eccentric muscle action appear to influence subsequent muscle performance more than isometric action, suggesting that dynamic exercise would have a greater effect on PAP than isometric activity.

Given the lack of evidence to support any superiority with either MVC or dynamic pre conditioning activities, it is likely that other consideration will determine which method to use. As the application of PAP via complex training methodologies will probably be a part of an overall training session, then it is likely that the PAP inducing activity also needs to contribute to the net training effect of the entire conditioning session. In this way heavy load exercises such as squats are likely to be preferred as these confer a training effect in themselves.

Where an optimally potentiated warm-up is the aim, then consistent access and availability of the methodology may be of greater concern. The use of MVC's may provide a viable option in these circumstances, where the logistics of modulating loads etc may prove a challenge to the application of the dynamic exercise. However, the use of potentiation in warm-up has yet to be validated, and the individuality of PAP response discussed later, adds a further layer of complexity to this application.

PAP is modulated by the volumes of the pre-conditioning stimulus,²⁶ and so, in addition to the nature of the PAP inducing exercise, another important element to consider will be the duration of the activity. As each activity will have both potentiating and fatiguing effects then the duration of exercise becomes a critical question. Hilfiker *et al.*²⁵ suggest that total contraction time should be less than 10 seconds to avoid undue fatigue. Again further research is needed to validate an optimal duration of PAP inducing activity, and here again the individual response characteristics will further complicate the quest. The vast majority of studies have used traditional "sets" of consecutive MVC's or resistance exercises to induce PAP, where the overall duration of the exercise is largely determined by the number of repetitions. Batsista *et al.*⁴ found that PAP could be elicited via intermittent exercise, and this may be another option whereby PAP can be elicited whilst controlling fatigue.

Determining the optimal load

If dynamic exercises such as squats are to be used as the pre conditioning activity, then a key variable will be the load utilised. As PAP is most associated with Type II muscle fibres,²³ then the pre-conditioning activity needs to ensure that an appropriate number of Type II

fibres are stimulated, whether through high resistance or high velocity. Verkoshansky⁵³ suggests a load of 90% 1 RM is utilised to optimise the development of PAP. The majority of current human studies have therefore utilised a high resistive load, whether in terms of isometric or concentric resistance as the preload stimulus to elicit PAP.¹² Unfortunately, again the results on the optimal load for eliciting PAP are equivocal.

While 5RM loads have been used in the majority of studies¹¹ the results from this resistance have been mixed. In a number of studies, significant increases in subsequent performances were found,^{17,39,56} while in others, no significant differences were noted.^{27,29,30,45} These differences could be due to the fatiguing effects of RM loads, which may in themselves require a longer recovery interval as the temporal profile of PAP will vary according to the intensity of exercise used to induce it.¹⁹

Research into other loads is sparse.¹² Baker³ found an increase in an explosive bench throw with a load of 65% of 1RM, and Yetter *et al.*⁵⁵ found an increase in running speed with loads of 30%, 50% and 70% of 1RM, suggesting that PAP can be induced with a relatively light load. Weber *et al.*⁵⁴ also found an increase in jump performance using 85% 1 RM loads, again suggesting that lighter loads may produce potentiation. However Hanson *et al.*²⁴ investigating the effects of squats at 40% 1RM and 80% 1 RM on repeated CMJ performance, found no change in either ground contact time, or peak ground reaction forces, suggesting that these loads were too light to produce potentiation.

Chiu *et al.*⁷ found an increase in PAP with a load of 90% of 1RM but only found significant differences in strength trained athletes. Additionally McBride *et al.*³⁵ and Chatzopoulos *et al.*⁶ found that a load of 90% was able to produce an improved 40 metres dash time³⁵ and 30 metre dash time⁶ although McBride³⁵ found no change at 10 and 30 metres. Mangus *et al.*³⁴ on the other hand found no potentiation effects on CMJ performance following squats at 90% 1 RM (although only 1 repetition was used which may have produced an insufficient stimulus). Bazett Jones *et al.*⁵ investigating the effects of 3 sets of 3 reps of leg press at 90% 1 RM on isometric squat performance showed no change in peak force and a significant decrease in rate of force development. Gilbert *et al.*¹⁹ found potentiation to be elicited at 100% of 1RM loads.

Clearly the evidence on optimal loads is highly equivocal and at present no definitive guidelines can be advised with which to optimise PAP. However, based on the physiological adaptations and the studies to date, it would appear that any potentiation is greatest where heavy loads are utilised.

The optimal time frame of PAP

In order to optimally utilise the effects of PAP on subsequent measures of athletic performance, it is important to optimise the time lag between the potentiating activity and subsequent performance. The traditional view has been that this time period should be minimal¹⁴ to maximise the heightened neural stimulation provided by the strength exercise. Initial recommendations for rest periods between exercises within a complex pair ranged from 30 seconds to five minutes.⁹ However, given the complex interactions between potentiation and fatigue, this thinking could

be flawed, and while potentiation may peak immediately post exercise, accumulated fatigue may mask this effect on subsequent performance. This balancing of PAP with fatigue has led to a wider variety of time intervals to be proposed.

Empirical research has investigated a much greater range of rest intervals varying between 10 seconds³⁰ and 20 minutes.²⁹ PAP has been demonstrated at timeframes from 1 minute²⁵ to 20 minutes.¹⁹ General evidence from these studies to date suggest that shorter rest periods (less than 4 minutes) are less effective than longer rest periods, although again the results are equivocal. Jensen and Ebben³⁰ investigating a range of time frames found non significant improvements in performance following 5RM squats at 1,2,3 and four minutes, while performance at 10 seconds was non-significantly reduced. They concluded that short rest periods are less effective at enhancing performance than longer rest periods.³⁰ This work supports the concept that potentiation and fatigue can co-exist, and that subsequent performance will depend upon the interaction between these factors, which is likely to depend upon both the characteristics of the pre-conditioning activity and the athlete. The fact that no timeframe is optimal is supported by Jones and Lees²⁹ who found no significant performance changes at 3, 10 and 20 minutes post 5RM squats, suggesting that neither long nor short time frames are optimal for PAP. Kilduff *et al.*³¹ investigating the effects of PAP in both the lower body (squats) and upper body (bench press) found potentiation at 8, 12 and 16 minutes for the lower body, and 4 8 and 12 minutes for the upper body, recommending a rest period of 8 minutes as the optimal timeframe for PAP. In this way the extent of any PAP produced will vary over time, as the balance between PAP and fatigue shifts.

The practical application of PAP will rely greatly on the timescale, and the current findings are ambiguous and do not provide the coach with the required degree of accuracy with which to develop optimal protocols. It seems likely that optimal PAP will only be elicited at a given window of opportunity and away from this performance could be impaired, have no impact on performance or show a limited degree of PAP. To further complicate the matter, Paasuke *et al.*³⁸ suggest that the timeframe of PAP will be related to both the training status of the athlete and the type of training undertaken. In measuring the decline in PAP following 10 second MVC's, differences were found between the endurance and power trained athletes in the rate of decay of PAP.

The impact of timeframe is clearly demonstrated by Chatzopoulos *et al.*⁶ who found no performance increases in sprint times 3 minutes subsequent to squats, but significant improvements 5 minutes post. This great variation in findings makes defining an optimal time frame for PAP impossible at present. Additionally, the suggestion that the optimal timeframe is highly individual¹⁰ adds to the challenge in optimally utilising PAP especially in group situations. If this is the case, then group applications of PAP may be limited by design, and PAP should be seen as an individually targeted training phenomenon.

Athlete differences

Twitch contractile studies in mammals have consistently demonstrated that PAP is at its greatest in Type II fibres.²³ Although in humans this correlation has

not been found consistently proved,²³ the findings of their study suggest that "human muscles with shorter twitch contraction times and a higher percentage of Type II fibres exhibit greater PAP".²³ Type II fibres undergo "greater phosphorylation of myosin regulatory light chains in a response to a conditioning activity, and may therefore have a greater capacity for PAP."⁴⁴ Thus, PAP is more likely to be evident in athletes with a higher percentage of Type II fibres, and within muscles with a higher percentage of Type II fibres.

Given the role of Type II fibres, and the net balance between potentiation and fatigue, the characteristics of the individual will likely play an important role in establishing the optimal method of inducing PAP. Studies have demonstrated that by that PAP is greater amongst stronger subjects^{7,8,13,20} and Stone *et al.*⁴⁹ suggest that strength levels may be an important predictor of PAP. Hamada *et al.*²³ in a study of triathletes, runners and active controls, demonstrated that PAP was greatest in trained muscles and where muscles were untrained PAP was not evident. Thus, the degree of training, relative to the activity to be enhanced is highly likely to influence the extent of PAP experienced. This is likely due to the ability of advanced athletes to "activate the high threshold fast motor units, whose muscle fibres exhibit the greatest PAP",⁴⁴ along with selective hypertrophy of Type II fibres in strength trained athletes, again increasing the potential for PAP. Sale⁴⁴ also suggests that training may alter light chain myosin composition, again increasing the potential for PAP. This also ties in with the probably higher levels of Type II fibres in stronger individuals, and the neural advantages these have for PAP. However these findings are not universal, and McBride (35) in a study on NCAA footballers found that PAP effects were independent of strength level. This conclusion was also supported by Mangus *et al.*,³⁴ who found that a strength ratio did not predict individuals likely to gain from PAP activities.

Similarly, the resistance to fatigue is likely to affect the effectiveness of the PAP, and will significantly affect the optimum time delay between conditioning activity and performance. Resistance to fatigue is again dependent on training status and so in the authors view PAP is again likely to be most effective at enhancing the performance of highly developed athletes as against untrained or recreationally trained athletes. This also produces a challenge to the individualisation of PAP methodologies, as it may be that as fatigue tolerance will vary over time. If this is the case then conceivably, the optimal timeframe of PAP may be an ever changing variable depending upon the condition and stress tolerance of an athlete at any specific time.

Logistical issues in eliciting PAP

There is an optimal timeframe when a muscle has recovered but is still potentiated,²⁶ at which time power based performance can be enhanced. A major challenge for the strength and conditioning professional is that it appears that this time frame will vary between individuals.^{10,22} In this way, optimising group training protocols may be highly challenging if not impossible, and the optimal application of PAP may only be possible on an individual prescription basis.

The application of PAP must also be evaluated in term of its importance within the overall conditioning process. The coach will need to carefully evaluate any potential benefit to be gained from the application of

PAP methodologies in either warm-up procedures or complex training methodologies. While PAP may be a very real physiological phenomenon, the elusive chase to apply its potential may require practices whereby training productivity from other elements of the session may be lost. Therefore, the overall effectiveness of any training session or warm-up must be the determining factor in whether or not to utilise PAP methodologies rather than simply being able to elicit PAP. The search for PAP to the overall detriment to an athlete's training programme is not to be commended.

In terms of warm-up, the application of individualised PAP inducing methodologies applied at the precise timeframe and load which optimises PAP may have performance enhancement potential, but may prove impossible to apply. This is especially the case in a team environment, where a large number of individual optimal applications are likely to be present. Here the application of the PAP protocol may result in problems with other elements of the performance preparation. While this may be more feasible for individual athletes, this again assumes the availability of the equipment required for eliciting PAP at the event location, and at the time required. While anecdotal stories exist about the application of heavy squats prior to elite level performance, an examination of the logistics of this question its applicability, given the timeframe of athletes in holding areas etc.

Even within complex training sessions, the timeframes of application may negatively affect other elements of a session. For example, where a rest interval of 8 minutes is recommended between a squat and jump, this would naturally result in a very low volume of work being performed in a given timescale, which over time could outweigh any potential potentiating benefits of the complex training protocol. Again a careful costs benefit analysis needs to be carried out to determine whether or not to utilise this phenomenon.

An area that could prove beneficial is the manipulation of loads on subsequent sets of resistance training exercises, where light loads are preceded by heavy loads. This system of strength-power-potentiating complexes was shown to produce significant increases in peak velocity and non significant increases in peak force, peak power and RFD in elite weightlifters.⁴⁹ This manipulation of loads in a contrast method may provide for an effective utilisation of PAP during resistance training sessions, but more work needs to be carried out prior to its application across training groups. It may also be an effective warm-up tool for weightlifting competitions, where heavy pulls or similar exercises could be used as a potentiating activity.⁴⁹

Long term studies

In terms of the long term benefit to be accrued from being better able to utilise the physiological mechanisms of PAP, the jury is still out. At present no longitudinal studies have examined the effects of complex training on either performance or neuromuscular adaptations. In this way, no decisions can be made on the long term advantages of complex training protocols over more traditional resistance training and plyometric protocols.

General conclusions and recommendations

Given the mixed evidence to date on PAP, the following recommendations can be given to assist coaches in the

application of this phenomenon.

- 1) PAP is a very real physiological phenomenon, but its effect on sports performance remains equivocal.
- 2) The optimal application of PAP will likely be highly individual, and require research and application on an individual basis.
- 3) MVC's and heavy resistance training exercises offer the opportunity to elicit PAP. In light of current research, neither would appear to have significant advantages over the other, and decisions as to which protocol should be used will be based upon other programming issues together with any logistical issues such as access etc.
- 4) While no definitive optimal load can be identified, loads will need to directly affect Type II muscle fibres, and thus will need to be high (>90% 1RM).
- 5) The optimal rest period will depend upon the interaction between potentiation and fatigue. This is likely to be highly individual and depend upon the pre conditioning load and activity, as well as the individual's fatigue tolerance, and potentiation potential. This may also change over time in response to the stress tolerance and potentiation potential etc of an individual at any given time.
- 6) It is likely that athletes with a high training age, a significant strength potential and/or a high percentage of Type II fibres are likely to be the highest responders to PAP.
- 7) As PAP is greatest in trained athletes, it should be seen as an advanced training tool, and used with experienced athletes.
- 8) While PAP is likely a physiological phenomena, it should not be chased at the expense of other training aims. If the effectiveness of other training methodologies has to be compromised to elicit optimal PAP then it's use must be subject to a rigorous cost benefit analysis.
- 9) Contrasting loads to provide for a strength-power potentiating complex during resistance training sessions may be an effective utilisation of PAP where movement velocity is an important factor.

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